

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 02 · Polynomials

Questions

Student book · questions with writing space

- 46 pages
- Past papers up to January 2026

PURE 2 | TOPIC 02

Polynomials

Your classified route

15

QUESTIONS

117

MARKS

4

FAMILIES

4

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 02 | FAMILY 01

Polynomial Division

WHAT THIS FAMILY ASKS

Recognise the polynomial division demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**2 classified questions**

19 total marks

4. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 + 13x^2 - 10x + 8$$

- (a) When $f(x)$ is divided by $(x - 2)$ the remainder is R and the quotient is $Q(x)$.
- (i) Find $Q(x)$.
- (ii) Find R .

(4)

Adaptive unit 1 · WMA12-01_2025_Oct_Q04

DR ESAM AHMED

4. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 + 13x^2 - 10x + 8$$

- (b) (i) Use the factor theorem to show that $(x + 4)$ is a factor of $f(x)$.
- (ii) Hence prove, using algebra, that the equation $f(x) = 0$ has only one real solution. **(5)**

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4. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 + 13x^2 - 10x + 8$$

(c) Find the range of values of x for which $f(x)$ is decreasing.

(3)

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3. (i) Given that
- $f(x) = 4x^3 + 6x + k$, where k is a constant
 - $(x + 2)$ is a factor of $f(x)$
- find the remainder when $f(x)$ is divided by $(x - 5)$

(4)

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DR ESAM AHMED

(ii) Find the remainder, R , and the quotient, $Q(x)$, when

$$6x^3 - 15x^2 - 21x + 8$$

is divided by $(2x + 3)$

(3)

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TOPIC 02 | FAMILY 02

Factor Theorem

WHAT THIS FAMILY ASKS

Recognise the factor theorem demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**7 classified questions**

58 total marks

3.

$f(x) = 6x^3 + 17x^2 + 4x - 12$

(a) Use the factor theorem to show that $(2x + 3)$ is a factor of $f(x)$.

(2)

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3.

$f(x) = 6x^3 + 17x^2 + 4x - 12$

(b) Hence, using algebra, write $f(x)$ as a product of three linear factors.

(4)

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$$f(x) = 6x^3 + 17x^2 + 4x - 12$$

(2)

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$$f(x) = (x^2 - 2)(2x - 3) - 21$$

(a) State the value of the remainder when $f(x)$ is divided by $(2x - 3)$

(1)

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4.

$f(x) = (x^2 - 2)(2x - 3) - 21$

(b) Use the factor theorem to show that $(x - 3)$ is a factor of $f(x)$

(2)

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$$f(x) = (x^2 - 2)(2x - 3) - 21$$

(5)

5.

$$f(x) = x^3 + (p + 3)x^2 - x + q$$

where p and q are constants and $p > 0$

Given that $(x - 3)$ is a factor of $f(x)$

(a) show that

$$9p + q = -51$$

(2)

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5.

$$f(x) = x^3 + (p + 3)x^2 - x + q$$

where p and q are constants and $p > 0$

Given that $(x - 3)$ is a factor of $f(x)$

Given also that when $f(x)$ is divided by $(x + p)$ the remainder is 9

(b) show that

$$3p^2 + p + q - 9 = 0$$

(2)

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5. $f(x) = x^3 + (p + 3)x^2 - x + q$
where p and q are constants and $p > 0$
Given that $(x - 3)$ is a factor of $f(x)$

(c) Hence find the value of p and the value of q .

(3)

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5. $f(x) = x^3 + (p + 3)x^2 - x + q$
where p and q are constants and $p > 0$
Given that $(x - 3)$ is a factor of $f(x)$

(d) Hence find a quadratic expression $g(x)$ such that

$$f(x) = (x - 3)g(x)$$

(2)

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4.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

$$f(x) = 4x^3 + ax^2 - 29x + b$$

where a and b are constants.

Given that $(2x + 1)$ is a factor of $f(x)$,

(a) show that

$$a + 4b = -56$$

(2)

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4. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 + ax^2 - 29x + b$$

where a and b are constants.

Given that $(2x + 1)$ is a factor of $f(x)$,

Given also that when $f(x)$ is divided by $(x - 2)$ the remainder is -25

(b) find a second simplified equation linking a and b .

(2)

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4. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 + ax^2 - 29x + b$$

where a and b are constants.

Given that $(2x + 1)$ is a factor of $f(x)$,

- (c) Hence, using algebra and showing your working,
- (i) find the value of a and the value of b ,
 - (ii) fully factorise $f(x)$.

(5)

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4.

$$f(x) = (x - 2)(2x^2 + 5x + k) + 21$$

where k is a constant.

(a) State the remainder when $f(x)$ is divided by $(x - 2)$

(1)

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4.

$f(x) = (x - 2)(2x^2 + 5x + k) + 21$

where k is a constant.

Given that $(2x - 1)$ is a factor of $f(x)$

(b) show that $k = 11$

(2)

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4. $f(x) = (x - 2)(2x^2 + 5x + k) + 21$
where k is a constant.

- (c) Hence
- (i) fully factorise $f(x)$,
 - (ii) find the number of real solutions of the equation

$$f(x) = 0$$

giving a reason for your answer.

(5)

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3.

$$f(x) = 2x^3 - x^2 + Ax + B$$

where A and B are integers.

Given that when $f(x)$ is divided by $(x + 3)$ the remainder is 55

(a) show that

$$3A - B = -118$$

(2)

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3. $f(x) = 2x^3 - x^2 + Ax + B$
where A and B are integers.
Given that when $f(x)$ is divided by $(x + 3)$ the remainder is 55

Given also that $(2x - 5)$ is a factor of $f(x)$,
(b) find the value of A and the value of B .

(3)

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3. $f(x) = 2x^3 - x^2 + Ax + B$
where A and B are integers.
Given that when $f(x)$ is divided by $(x + 3)$ the remainder is 55

(c) Hence find the quotient when $f(x)$ is divided by $(x - 7)$

(2)

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5.

$f(x) = 3x^3 + ax^2 - 10x + b$

where a and b are constants.

Given that $(3x - 4)$ is a factor of $f(x)$,

(a) show that $16a + 9b = 56$

(2)

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5. $f(x) = 3x^3 + ax^2 - 10x + b$
where a and b are constants.
Given that $(3x - 4)$ is a factor of $f(x)$,

Given further that when $f(x)$ is divided by $(x - 2)$ the remainder is b ,
(b) find the value of a and the value of b .

(4)

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5. $f(x) = 3x^3 + ax^2 - 10x + b$
where a and b are constants.
Given that $(3x - 4)$ is a factor of $f(x)$,

(c) Hence, using algebra, fully factorise $f(x)$.

(3)

Adaptive unit 3 · WMA12-01_2025_Jan_Q05

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TOPIC 02 | FAMILY 03

Remainder Theorem

WHAT THIS FAMILY ASKS

Recognise the remainder theorem demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**5 classified questions**

34 total marks

(a) State the remainder when $f(x)$ is divided by $(x - 3)$.

(1)

4. $f(x) = (x - 3)(3x^2 + x + a) - 35$ where a is a constant

Given $(3x - 2)$ is a factor of $f(x)$,

(b) show that $a = -17$

(2)

Adaptive unit 2 · WMA12-01_2019_Oct_Q04

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$f(x) = (x - 3)(3x^2 + x + a) - 35$ where a is a constant

(5)

Adaptive unit 3 · WMA12-01 2019 Oct Q04

1.

$$f(x) = x^4 + ax^3 - 3x^2 + bx + 5$$

where a and b are constants.

When $f(x)$ is divided by $(x + 1)$, the remainder is 4

(a) Show that $a + b = -1$

(2)

Adaptive unit 1 · WMA12-01_2021_Jan_Q01

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1. $f(x) = x^4 + ax^3 - 3x^2 + bx + 5$
where a and b are constants.
When $f(x)$ is divided by $(x + 1)$, the remainder is 4

When $f(x)$ is divided by $(x - 2)$, the remainder is -23

(b) Find the value of a and the value of b .

(4)

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5.

$$f(x) = 3x^3 + Ax^2 + Bx - 10$$

where A and B are integers.

Given that

- when $f(x)$ is divided by $(x - 1)$ the remainder is k
- when $f(x)$ is divided by $(x + 1)$ the remainder is $-10k$
- k is a constant

(a) show that

$$11A + 9B = 83$$

(3)

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5. $f(x) = 3x^3 + Ax^2 + Bx - 10$
where A and B are integers.
Given that
- when $f(x)$ is divided by $(x - 1)$ the remainder is k
 - when $f(x)$ is divided by $(x + 1)$ the remainder is $-10k$
 - k is a constant

Given also that $(3x - 2)$ is a factor of $f(x)$,
(b) find the value of A and the value of B .

(3)

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5. $f(x) = 3x^3 + Ax^2 + Bx - 10$

where A and B are integers.

Given that

- when $f(x)$ is divided by $(x - 1)$ the remainder is k
- when $f(x)$ is divided by $(x + 1)$ the remainder is $-10k$
- k is a constant

(c) Hence find the quadratic expression $g(x)$ such that

$$f(x) = (3x - 2)g(x)$$

(2)

Adaptive unit 3 · WMA12-01_2022_Jan_Q05

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1.

$f(x) = ax^3 + 3x^2 - 8x + 2$ where a is a constant

Given that when $f(x)$ is divided by $(x - 2)$ the remainder is 3, find the value of a .

(3)

Adaptive unit 1 · WMA12-01_2024_Jan_Q01

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4: The function $f(x)$ is defined by

$$f(x) = ax^3 + bx^2 + 5x - 3$$

where a and b are constants.

Given that $(x + 3)$ is a factor of $f(x)$,

(a) show that

$$b = 3a + 2$$

(2)

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4: The function $f(x)$ is defined by

$$f(x) = ax^3 + bx^2 + 5x - 3$$

where a and b are constants.

Given that $(x + 3)$ is a factor of $f(x)$,

Given further that when $f(x)$ is divided by $(2x - 1)$ the remainder is $\frac{7}{4}$

(b) find the value of a and the value of b .

(4)

Adaptive unit 2 · WMA12-01_2025_MayJune_Q04

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4: The function $f(x)$ is defined by

$$f(x) = ax^3 + bx^2 + 5x - 3$$

where a and b are constants.

Given that $(x + 3)$ is a factor of $f(x)$,

(c) Using algebra, find the quotient and the remainder when $f(x)$ is divided by $(x - 2)$

(3)

Adaptive unit 3 · WMA12-01_2025_MayJune_Q04

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TOPIC 02 | FAMILY 04

Roots/Factors

WHAT THIS FAMILY ASKS

Recognise the roots/factors demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

1 classified questions

6 total marks

2.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

$$f(x) = 4x^3 - 8x^2 + 5x + a$$

where a is a constant.

Given that $(2x - 3)$ is a factor of $f(x)$,

(a) use the factor theorem to show that $a = -3$

(2)

Adaptive unit 1 · WMA12-01_2023_MayJune_Q02

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2. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 - 8x^2 + 5x + a$$

where a is a constant.
Given that $(2x - 3)$ is a factor of $f(x)$,

(b) Hence show that the equation $f(x) = 0$ has only one real root.

(4)

Adaptive unit 2 · WMA12-01_2023_MayJune_Q02

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